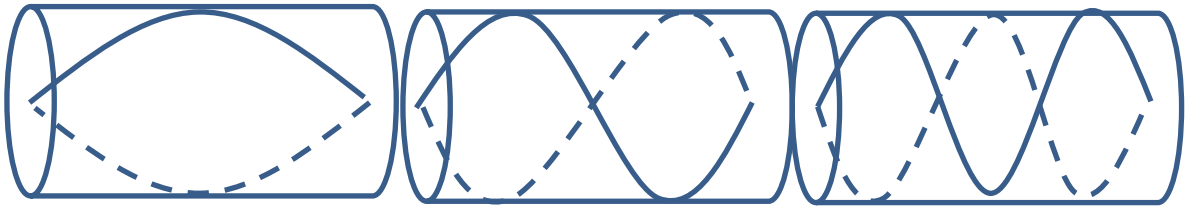
LESSON 7

CIRCULAR WAVEGUIDE RESONATOR

7.1 Introduction

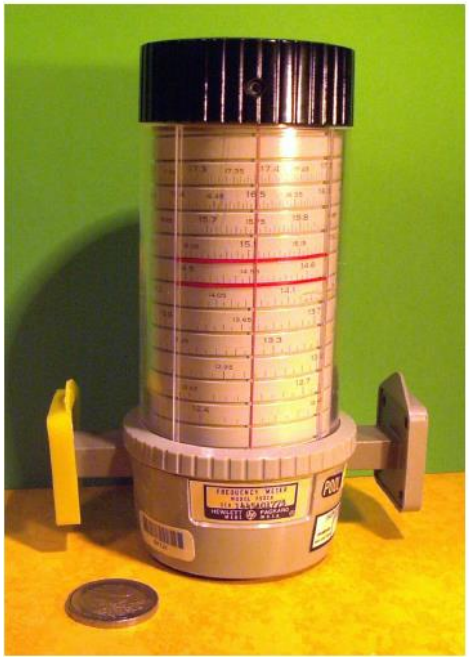
-When one end of the waveguide is terminated with a shorting plate, there will be reflections causing standing waves.

-When another shorting plate is kept at a distance of multiples of $\lambda g/2$, then the hollow space so formed can support a signal that bounces back and forth between the two shorting plates. This results in resonance.

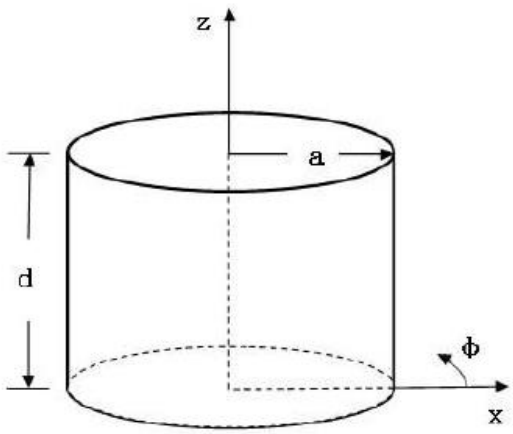


-The hollow space is called cavity and the arrangement so done is called **cavity resonator.**

-Resonators are mainly used in storage, klystron tubes, band pass filters, microwave cookers, waveguide meter. (power supply, waveguide feed, cavity).



7.2 Circular waveguide resonator



We know that the general equations governing propagation of electromagnetic wave in a circular waveguide are

$$\frac{\partial^2 E_z}{\partial r^2} + \frac{1}{r}\frac{\partial E_z}{\partial r} + \frac{1}{r^2}\frac{\partial^2 E_z}{\partial \theta^2} + \frac{\partial^2 E_z}{\partial z^2} = -\omega^2 \mu \varepsilon E_z \qquad (6.1)$$

And

$$\frac{\partial^2 H_z}{\partial r^2} + \frac{1}{r}\frac{\partial H_z}{\partial r} + \frac{1}{r^2}\frac{\partial^2 H_z}{\partial \theta^2} + \frac{\partial^2 H_z}{\partial z^2} = -\omega^2 \mu \varepsilon H_z \qquad (6.2)$$

For TE mode the solution to eq. 6.2 is

$$H_z = CJ_n'(\mathrm{k}_c\mathrm{r})[Asin(n\theta)]e^{i(\omega t - kz)} \qquad (6.3)$$

For TM mode the solution to eq. 6.1 is

$$E_z = CJ_n(k_c r)[Asin(n\theta)]e^{i(\omega t - kz)} \qquad (6.4)$$

7.3 Expressions for r and θ field components in a circular waveguide,
These are

$$E_r = \frac{-i}{k_c^2}\left(k\frac{\partial E_z}{\partial r} - \frac{\omega\mu}{r}\frac{\partial H_z}{\partial \theta}\right) \qquad \text{c} \qquad (2.33)$$

$$H_\theta = \frac{-i}{k_c^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial r} - \frac{k}{r}\frac{\partial H_z}{\partial \theta}\right) \qquad \text{c} \qquad (2.34)$$

$$E_\theta = \frac{-i}{k_c^2}\left(\frac{k}{r}\frac{\partial E_z}{\partial \theta} - \omega\mu\frac{\partial H_z}{\partial r}\right) \qquad \text{c} \qquad (2.35)$$

$$H_r = \frac{i}{k_c^2}\left(\frac{\omega\varepsilon}{r}\frac{\partial E_z}{\partial\theta} - k\frac{\partial H_z}{\partial r}\right) \tag{2.36}$$

Where $k_c^2 = \omega^2\mu\varepsilon - k^2$.

7.4 Expressions for TM_{nm} field components in a circular waveguide,
Substituting for $E_z = CJ_n(k_c r)[Asin(n\theta)]e^{i(\omega t - kz)}$ and $H_z = 0$, the field components for TM_{mn} modes are

$$E_z = CJ_n(k_c r)[Asin(n\theta)]e^{i(\omega t - kz)}$$

$$H_z = 0$$

$$E_r = \frac{-i}{k_c^2}\left(k\frac{\partial E_z}{\partial r}\right) = -\frac{ik}{k_c}CJ_n^*(k_c r)[Asin(n\theta)]e^{i(\omega t - kz)}$$

(2.33)

$$E_\theta = \frac{-i}{k_c^2}\left(\frac{k}{r}\frac{\partial E_z}{\partial\theta}\right) = -\frac{ikn}{k_c^2 r}CJ_n(k_c r)[Acos(n\theta)]e^{i(\omega t - kz)}$$

$$H_r = \frac{i}{k_c^2}\left(\frac{\omega\varepsilon}{r}\frac{\partial E_z}{\partial\theta}\right) = \frac{\omega\varepsilon n}{k_c^2 r}CJ_n(k_c r)[Acos(n\theta)]e^{i(\omega t - kz)}$$

$$H_\theta = \frac{-i}{k_c^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial r}\right) = -\frac{i\omega\varepsilon}{k_c}CJ_n^*(k_c r)[Asin(n\theta)]e^{i(\omega t - kz)}$$

(2.34)

where * denotes derivative with respects to r.

7.5 Expressions for TE_{nm} field components in a circular waveguide,
Substituting for $E_z = 0$ and, $H_z = CJ_n'(k_c r)[A\sin(n\theta) + B\cos(n\theta)]\, e^{i(\omega t - \beta z)}$ the field components for TE_{mn} modes are

$$E_r = \frac{-i}{k_c^2}\left(k\frac{\partial E_z}{\partial r} - \frac{\omega\mu}{r}\frac{\partial H_z}{\partial\theta}\right) = \frac{i\omega\mu n}{k_c^2 r}CJ_n'(k_c r)[A\cos(n\theta)]\, e^{i(\omega t - kz)}$$

(2.37)

$$E_\theta = \frac{-i}{k_c^2}\left(\frac{k}{r}\frac{\partial E_z}{\partial\theta} - \omega\mu\frac{\partial H_z}{\partial r}\right) = \frac{i\omega\mu}{k_c}CJ_n'^*(k_c r)[A\sin(n\theta)]\, e^{i(\omega t - kz)}$$

$$E_z = 0$$

$$H_r = \frac{i}{k_c^2}\left(\frac{\omega\varepsilon}{r}\frac{\partial E_z}{\partial\theta} - k\frac{\partial H_z}{\partial r}\right) = -\frac{ik}{k_c}CJ_n'^*[A\sin(n\theta)]\, e^{i(\omega t - kz)}$$

$$H_\theta = \frac{-i}{k_c^2}\left(\omega\varepsilon\frac{\partial E_z}{\partial r} - \frac{k}{r}\frac{\partial H_z}{\partial\theta}\right) = \frac{ikn}{k_c^2 r}CJ_n'(k_c r)[A\cos(n\theta)]\, e^{i(\omega t - kz)}$$

(2.38)

$$H_z = CJ_n'(k_c r)[A\sin(n\theta)]\, e^{i(\omega t - kz)} \tag{2.39}$$

7.6 TM modes in circular waveguide resonator

Consider the electric field components for TM mode in a circular waveguide

$$E_z = CJ_n(k_c r)[Asin(n\theta)]e^{i(\omega t - kz)} = De^{-ikz}$$

where $D = CJ_n(k_c r)[Asin(n\theta)]e^{i(\omega t)}$

$$E_r = -\frac{ik}{k_c} CJ_n^*(k_c r)[Asin(n\theta)]e^{i(\omega t - kz)} = -iEe^{-ikz}$$

where $E = \frac{k}{k_c} CJ_n^*(k_c r)[Asin(n\theta)]e^{i(\omega t)}$

$$E_\theta = -i\frac{kn}{k_c^2} CJ_n(k_c r)[Acos(n\theta)]e^{i(\omega t - kz)} = -iFe^{-ikz}$$

where $F = \frac{kn}{k_c^2} CJ_n(k_c r)[Acos(n\theta)]e^{i(\omega t)}$

After bouncing of the walls standing waves are formed giving

$$E_z = De^{-ikz} + De^{ikz} = D\left(e^{ikz} + e^{-\boldsymbol{ikz}}\right) = 2Dcos(kz)$$

$$E_z = 2CJ_n(k_c r)[Asin(n\theta)]cos(kz) \qquad (6.8a)$$

$$E_r = -iEe^{-ikz} + iEe^{ikz} = iE\left(e^{ikz} - e^{-\boldsymbol{ikz}}\right) = -2Esin(kz)$$

$$E_r = -\frac{2k}{k_c} CJ_n^*(k_c r)[Asin(n\theta)]sin(kz) \qquad (6.8b)$$

$$E_\theta = -iFe^{-ikz} + iFe^{ikz} = Fi\left(e^{ikz} - e^{-ikz}\right) = 2Fsin(kz)$$

$$E_\theta = -\frac{2kn}{k_c^2 r} CJ_n(k_c r)[Acos(n\theta)]sin(kz) \qquad (3)$$

H_θ and H_r can be found by using Maxwell's curl eq. for E in component form (note: $\partial/\partial r = \partial/\partial\theta = 0$ but $\partial/\partial z \neq 0$) $\quad \nabla \times \vec{E} = -i\omega\mu\vec{H}$

$$\left(\frac{1}{r}\frac{\partial E_z}{\partial\theta} - \frac{\partial E_\theta}{\partial z}\right)\hat{r} + \left(\frac{\partial E_r}{\partial z} - \frac{\partial E_z}{\partial r}\right)\hat{\theta} + \frac{1}{r}\left(\frac{\partial(rE_\theta)}{\partial r} - \frac{\partial E_r}{\partial\theta}\right)\hat{z} = -i\omega\mu(H_r\hat{r} + H_\theta\hat{\theta} + H_z\hat{z})$$

$$H_r = \frac{i}{\omega\mu}\left(\frac{1}{r}\frac{\partial E_z}{\partial\theta} - \frac{\partial E_\theta}{\partial z}\right) = -\frac{i}{\omega\mu}\frac{\partial}{\partial z}\left\{-\frac{2kn}{k_c^2 r} CJ_n(k_c r)[Acos(n\theta)]\sin(kz)\right\}$$

$$H_r = \frac{i}{\omega\mu}\frac{2k^2n}{k_c^2r}\ CJ_n(k_cr)[Acos(n\theta)]\cos(kz) \qquad (4)$$

$$H_\theta = \frac{i}{\omega\mu}\left(\frac{\partial E_r}{\partial z} - \frac{\partial E_z}{\partial r}\right) = \frac{i}{\omega\mu}\frac{\partial}{\partial z}\left\{-\frac{2k}{k_c}CJ_n^*(k_cr)[Asin(n\theta)]sin(kz)\right\}$$

$$H_\theta = -\frac{i}{\omega\mu}\frac{2k^2}{k_c}CJ_n^*(k_cr)[Asin(n\theta)]cos(kz) \qquad (5)$$

Furthermore $E_r = E_\theta = 0$ at z = d, implying that

$$E_r = -\frac{2k}{k_c}CJ_n^*(k_cr)[Asin(n\theta)]sin(kz) = 0 \qquad \text{if } k = \frac{p\pi}{d},$$

Where p=0,1,2,3,...

and

$$E_\theta = -\frac{2kn}{k_c^2r}\ CJ_n(k_cr)[Acos(n\theta)]sin(kz) = 0 \qquad \text{if } k = \frac{p\pi}{d}$$

Where p=0,1,2,3,...

Hence

$$E_z = 2CJ_n(k_cr)[Asin(n\theta)]cos(\frac{p\pi z}{d}) \qquad (5)$$

The propagation constant of the TM_{nm} mode is, from

$$\beta = k = \sqrt{\omega^2\mu\varepsilon - \left(\frac{X_{nm}}{a}\right)^2} = \sqrt{(\omega\sqrt{\varepsilon\mu})^2 - \left(\frac{X_{nm}}{a}\right)^2} \qquad (6)$$

Where X_{nm} is the root. Substituting $k = \frac{p\pi}{d}$ in eq.6, the resonant frequency is

$$\frac{p\pi}{d} = \sqrt{\omega^2\mu\varepsilon - \left(\frac{X_{nm}}{a}\right)^2} \qquad \text{or} \qquad f_{rnml} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{X_{nm}}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2}$$

5.2.2 TE mode in circular waveguide resonator

We know

$$H_z = CJ_n'(k_cr)[A\cos(n\theta)]e^{i(\omega t - kz)} = Fe^{-ikz} \qquad \text{and } E_z = 0$$

Where $F = CJ_n'(k_cr)[A\cos(n\theta)]e^{i(\omega t)}$

Consider the electric field components for TE modes in a circular waveguide

$$E_r = \frac{i\omega\mu n}{k_c^2 r} C J_n'(k_c r)[A\cos(n\theta)]e^{i(\omega t - kz)} = iDe^{-ikz}$$

where $D = \frac{\omega\mu n}{k_c^2 r} C J_n'(k_c r)[A\cos(n\theta)]e^{i(\omega t)}$

$$E_\theta = \frac{i\omega\mu}{k_c} C J_n'^*(k_c r)[A\sin(n\theta)]e^{i(\omega t - kz)} = iEe^{-ikz}$$

where $E = \frac{\omega\mu}{k_c} C J_n'^*(k_c r)[A\sin(n\theta)]e^{i(\omega t - kz)}$

After bouncing of the walls standing waves are formed giving

$$H_z = Fe^{-ikz} + Fe^{ikz} = F\left(e^{ikz} + e^{-ikz}\right) = 2Fcos(kz)$$

$$H_z = CJ_n'(k_c r)[A\cos(n\theta)]\cos(kz)$$

$$E_r = iDe^{-i\beta z} - iDe^{i\beta z} = -iD\left(e^{i\beta z} - e^{-i\beta z}\right) = 2Dsin(\beta z)$$

$$E_r = \frac{2\omega\mu n}{k_c^2 r} CJ_n'(k_c r)[A\cos(n\theta)]sin(\beta z) \qquad (1)$$

$$E_\theta = iEe^{-i\beta z} - iEe^{i\beta z} = -iE\left(e^{i\beta z} - e^{-i\beta z}\right) = 2Esin(\beta z)$$

$$E_\theta = \frac{2\omega\mu}{k_c} CJ_n'^*(k_c r)[A\sin(n\theta)]sin(\beta z) \qquad (2)$$

H_θ and H_r can be found by using Maxwell's curl eq. for E in component form

$$H_r = \frac{i}{\omega\mu}\left(\frac{1}{r}\frac{\partial E_z}{\partial\theta} - \frac{\partial E_\theta}{\partial z}\right) = -\frac{i}{\omega\mu}\frac{\partial}{\partial z}\left\{\frac{2\omega\mu}{k_c} CJ_n'^*(k_c r)[A\sin(n\theta)]sin(kz)\right\}$$

$$H_r = \frac{i}{\omega\mu}\frac{2\omega\mu}{k_c} CJ_n'^*(k_c r)[A\sin(n\theta)]cos(kz) \qquad (3)$$

$$H_\theta = \frac{i}{\omega\mu}\left(\frac{\partial E_r}{\partial z} - \frac{\partial E_z}{\partial r}\right) = \frac{i}{\omega\mu}\frac{\partial}{\partial z}\left\{\frac{2\omega\mu n}{k_c^2 r} CJ_n'(k_c r)[A\cos(n\theta)]sin(kz)\right\}$$

$$H_\theta = -\frac{i}{\omega\mu}\frac{2k\omega\mu n}{k_c^2 r} CJ_n'(k_c r)[A\cos(n\theta)]cos(kz) \qquad (4)$$

Furthermore $E_r = E_\theta = 0$ at z = d, implying that

$$E_r = \frac{2\omega\mu n}{k_c^2 r} CJ_n'(k_c r)[A\cos(n\theta)]sin(kz) \quad = 0 \qquad \text{if } k = \frac{p\pi}{d},$$

[7]

and

$$E_\theta = \frac{2\omega\mu}{k_c} C J_n'^{*}(k_c r)[A\sin(n\theta)]sin(kz) = 0 \qquad \text{if } k = \frac{p\pi}{d},$$

Since H_z cannot be equal to zero at z=d, we express it as

$$H_z = C J_n'(k_c r)[A\sin(n\theta)]sin(kz)\, e^{i(\omega t - kz)} \qquad (5)$$

The propagation constant of the TM_{nm} mode is, from

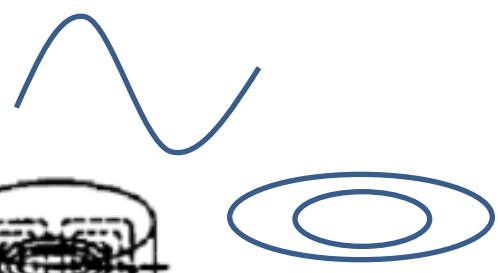
$$\beta = k = \sqrt{k^2 - \left(\frac{X_{nm}'}{a}\right)^2} = \sqrt{(\omega\sqrt{\varepsilon\mu})^2 - \left(\frac{X_{nm}'}{a}\right)^2} \qquad (6)$$

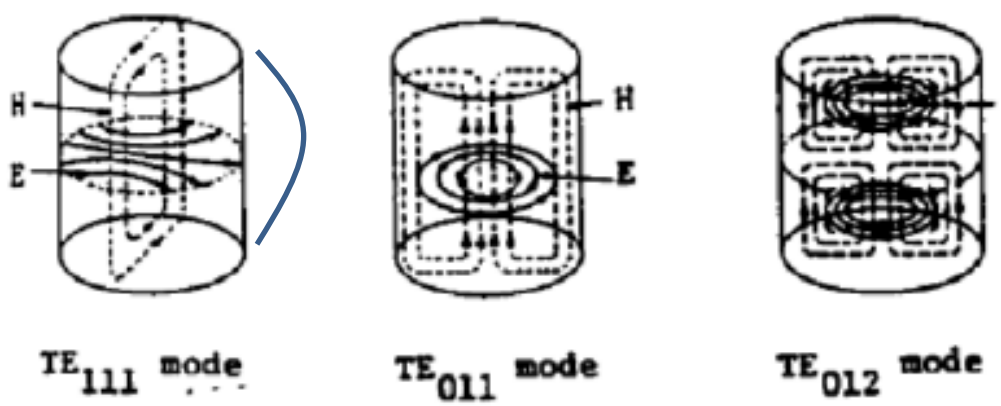
where X_{nm} is the root. Substituting $k = \frac{p\pi}{d}$ in eq.6, the resonant frequency is

$$\frac{p\pi}{d} = \sqrt{\omega^2\mu\varepsilon - \left(\frac{X_{nm}'}{a}\right)^2} \qquad \text{or} \qquad f_{rnml} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{X_{nm}'}{a}\right)^2 + \left(\frac{p\pi}{d}\right)^2} \qquad (7)$$

5.2.4 Field patterns for TE_{mnp} modes in a circular waveguide resonator

m=number of full sinusoids in the circumferential direction
n=number of half sinusoids in the radial direction
p=number of half sinusoids in the vertical (axial) direction





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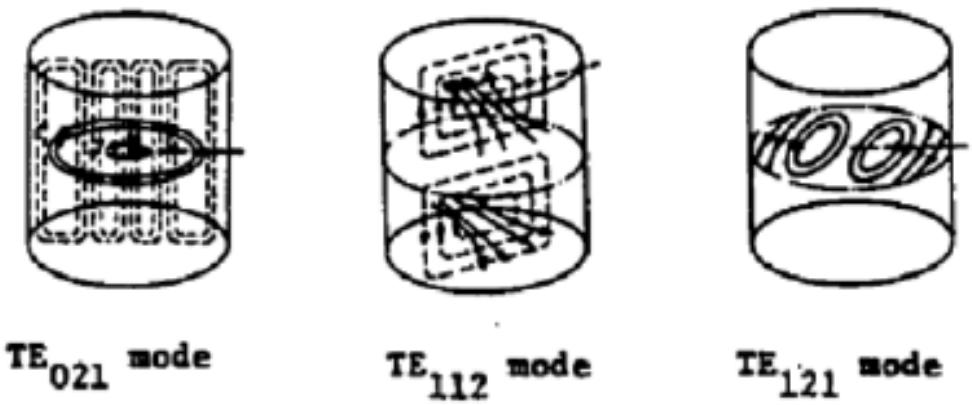
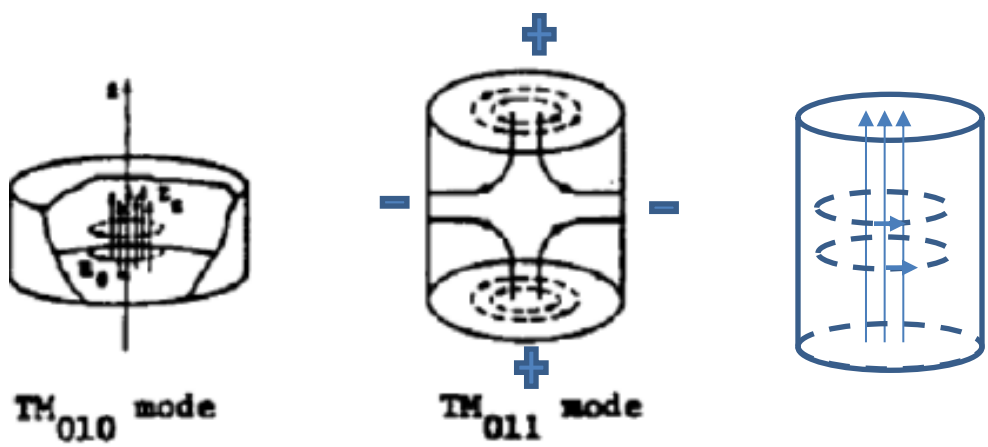


5.2.5 Field patterns for TM_{mnp} modes in circular waveguide resonator

m=number of full sinusoids in the circumferential direction
n=number of half sinusoids in the radial direction
p=number of half sinusoids in the vertical (Z)direction



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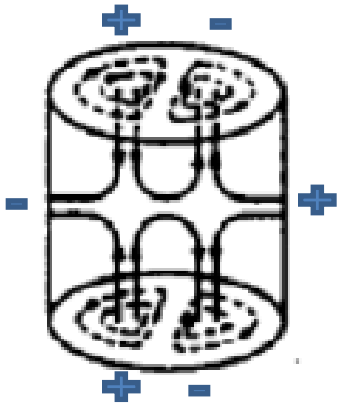
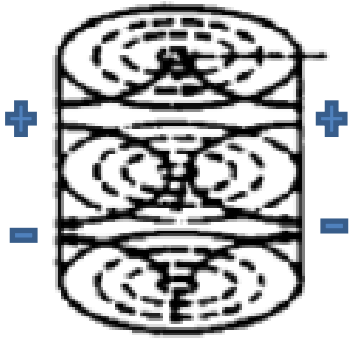
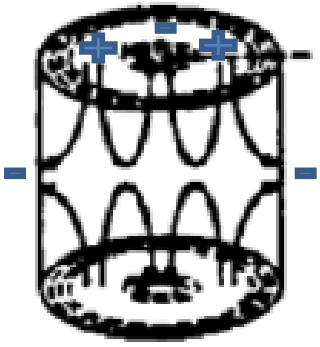


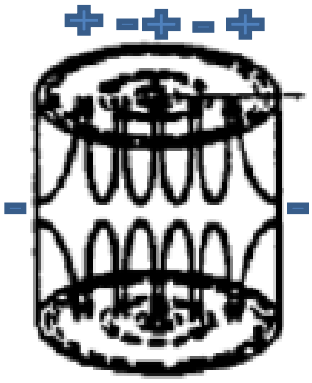
TM_{111} mode



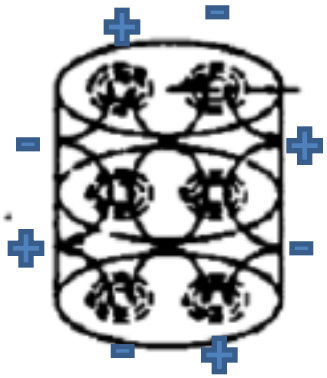
TM_{012} mode



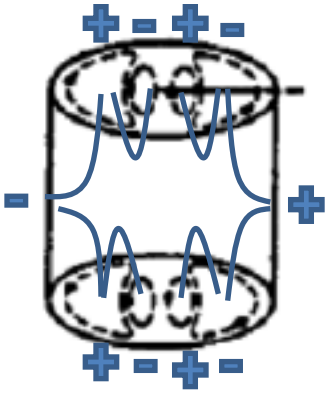
TM_{021} mode



TM_{031} mode



TM_{112} mode



TM_{121} mode